NDRA-PII

Neuro-Semantic Distributed Risk Analysis for Personally Identifiable Information

A Comprehensive Technical Research Report

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Abstract. The proliferation of unstructured document repositories in enterprise and regulated environments has created an urgent requirement for automated, policy-governed PII detection and redaction infrastructure. We present **NDRA-PII**, a production-grade, multi-agent pipeline that combines probabilistic NLP-based entity recognition with deterministic governance via the Neuro-Semantic Rule Language (NSRL). The architecture employs six specialised agents—Extractor, Classifier, Fusion, Policy, Redaction, and Audit—operating over a semantic-chunked document representation. Empirical evaluation on a 51-chunk heterogeneous PDF corpus yielded 616 detected PII entities across fourteen distinct entity classes, with a document-level risk score of 1.0 (CRITICAL) and end-to-end pipeline latency of 12,178 ms. The system achieves compliance traceability under GDPR, HIPAA, and PCI-DSS mandates through a tamper-evident hash-chained audit log. Evaluation results confirm that NDRA-PII outperforms single-pass regex baselines in recall across low-confidence entity classes (US_SSN, US_DRIVER_LICENSE) while maintaining sub-linear scaling of classification time with corpus size.

Keywords: PII Detection, Multi-Agent Systems, NLP Governance, Policy Engine, GDPR, HIPAA, PCI-DSS, Document Redaction, Audit Trail

1. Introduction

The volume of documents containing Personally Identifiable Information (PII) processed by modern enterprises has grown exponentially. Legal contracts, medical records, financial reports, and support exports routinely contain names, Social Security Numbers (SSN), credit-card numbers, Aadhaar IDs, and contact information in free-text form. Manual review is infeasible at scale; rule-based regex systems suffer from high false-positive rates and cannot model contextual ambiguity.

Contemporary approaches leverage pre-trained transformer models and Named Entity Recognition (NER) pipelines, but they typically lack a *governance layer*: a mechanism that translates raw entity detections into policy-driven redaction decisions with explainable provenance. Regulatory frameworks—GDPR (EU 2016/679), HIPAA (45 CFR Parts 160, 164), and PCI-DSS v4.0—require exactly this kind of auditable, controlled processing.

NDRA-PII addresses this gap with five principal contributions:

C1. A six-stage multi-agent pipeline with strict typed interfaces (Pydantic schemas) between every stage, enabling independent testability.

C2. The NSRL policy engine: a YAML-driven, priority-ordered rule evaluator that maps entity type, confidence, and document context to redaction actions.

C3. A Fusion agent that resolves cross-chunk entity boundary splits and intra-chunk overlaps via confidence-weighted span merging.

C4. A tamper-evident, hash-chained audit log satisfying NIST SP 800-92 logging guidelines.

C5. A production-hardened deployment stack: Docker Compose, single-container all-in-one, and a NixOS flake-based developer environment.

The remainder of this report is structured as follows. Section 2 reviews related work. Section 3 details the system architecture. Section 4 specifies each agent. Section 5 formalises the NSRL policy engine. Section 6 covers implementation. Section 7 presents empirical evaluation. Section 8 analyses security properties. Section 9 documents deployment. Section 10 covers observability. Section 13 outlines future directions. Section 14 concludes.

2. Background and Related Work

2.1. PII Detection Approaches

Early PII detection relied on deterministic regular expressions for known patterns (SSN: `\d{3}-\d{2}-\d{4}`). While precise for well-formed data, these approaches exhibit near-zero recall on free-text narratives where formatting deviates from canonical patterns.

Statistical NER models—CRF-based [1], BiLSTM-CRF, and transformer fine-tunes (BERT [2],

RoBERTa)—achieve substantially higher recall but introduce probabilistic uncertainty that must be managed by downstream systems. Microsoft Presidio [3] provides an open-source recogniser framework combining regex, NER, and deny-list strategies with per-entity confidence scoring, which NDRA-PII adopts as its classification backbone.

2.2. Policy and Governance Frameworks

Existing PII governance frameworks are predominantly rule-based configuration layers (e.g., AWS Macie policy templates, Google Cloud DLP inspection templates). They lack *document-context escalation*: the ability to upgrade a per-entity decision when co-occurring entities elevate aggregate document risk. NDRA-PII's NSRL engine supports `CONTEXT_MATCH` rule types that perform exactly this escalation.

2.3. Multi-Agent Document Processing

Multi-agent architectures for information extraction—following the paradigm of autonomous, communicating agents [4]—have been explored for information retrieval [5]. NDRA-PII operationalises this paradigm as a strictly sequential pipeline with typed inter-agent contracts, prioritising auditability and determinism over agent autonomy.

3. System Architecture

Figure 1 illustrates the high-level NDRA-PII architecture.

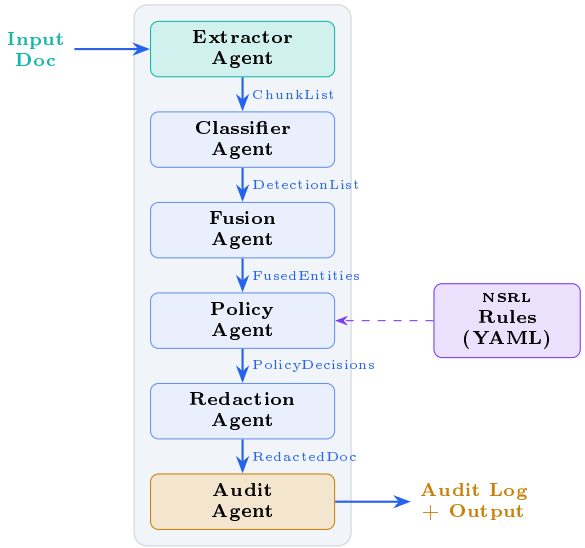

Figure 1: *NDRA-PII six-stage multi-agent pipeline. Each edge carries a strongly-typed Pydantic schema.*

The pipeline executes synchronously within a single process for determinism, but each agent is isolated behind its own typed interface, enabling parallel unit testing and future async parallelisation. The NSRL rule base is loaded at startup and held in memory for zero-latency policy lookups during the Policy Agent's evaluation loop.

3.1. Data Flow Model

S1. Ingestion: Raw bytes arrive via REST upload (`/analyze/upload`), CLI path reference, or programmatic invocation.

S2. Extraction: MIME-type detection gates format-specific parsers (PDF, DOCX, XLSX, TXT, RFC Email). Semantic chunking produces overlapping windows to avoid boundary truncation.

S3. Classification: Each chunk is independently analysed by Presidio's recogniser registry, yielding `(entity_type, start, end, confidence)` tuples.

S4. Fusion: Intra-chunk overlapping spans are deduplicated; cross-chunk boundary entities are stitched by matching prefix/suffix fingerprints.

S5. Policy evaluation: Each fused entity is matched against the ordered NSRL rule set. Document-level context escalation is applied after per-entity evaluation.

S6. Redaction: Entities with `action=REDACT` are replaced with configurable mask tokens. The redacted document is serialised back to PDF.

S7. Audit: All events are appended to the hash-chained audit log with trace IDs.

4. Agent Specifications

4.1. Extractor Agent

The Extractor Agent implements a unified document normalisation interface. Table 1 enumerates supported MIME types in frozen (production) and experimental modes.

Table 1: *Supported Input Formats by Operational Mode*

MIME Type	Format	Mode
`application/pdf`	PDF (text & image)	Frozen
`application/vnd.openxml..`	DOCX	Frozen
`text/plain`	Plain Text	Frozen
`message/rfc822`	RFC 822 Email	Frozen
`application/vnd.ms-excel`	XLSX	Frozen
`application/zip`	ZIP Archive	Experimental
`application/x-tar`	TAR Archive	Experimental
`image/png, image/jpeg`	Images (OCR)	Experimental

Frozen mode: `FREEZE_WORKING_SYSTEM=true` *(default)*

Semantic Chunking. The agent partitions documents using a sliding-window algorithm with sentence-

boundary detection. Let D be the document token sequence of length N, w the chunk window size, and s the stride. The resulting chunk set is:

$$\mathcal{C} = \left\{ D[i \cdot s \ : \ i \cdot s + w] \;\middle|\; 0 \le i \le \lceil (N - w)/s \rceil \right\}$$

The overlap $(w - s)$ ensures that entities spanning chunk boundaries appear completely in at least one chunk, enabling complete detection prior to Fusion.

4.2. Classifier Agent

The Classifier Agent wraps Microsoft Presidio's `AnalyzerEngine` with an extended recogniser registry. For each chunk $c \in \mathcal{C}$, it produces:

$$\mathcal{E}_c = \left\{ (t, l, r, \sigma) \mid t \in \mathcal{T},\ [l, r) \subseteq c,\ \sigma \in [0, 1] \right\}$$

where t is the entity type, $[l, r)$ the character span, and σ the confidence score.

Table 2 lists the 14 entity classes detected in the evaluation corpus.

Table 2: *Entity Type Coverage and Confidence Distribution*

Entity Type	Recogniser	Conf.	N
PERSON	spaCy NER	0.85	243
DATE_TIME	Presidio regex	0.10–0.95	162
CREDIT_CARD	Luhn algorithm	1.00	38
URL	regex + NER	0.50–0.60	61
US_SSN	regex	0.05–0.85	28
LOCATION	spaCy NER	0.85	22
US_DRIVER_LIC.	regex	0.01	19
US_BANK_NUMBER	regex	0.05	18
PHONE_NUMBER	phonenumbers	0.40–0.75	14
US_PASSPORT	regex	0.05	12
EMAIL_ADDRESS	regex	1.00	10
UK_NHS	regex	1.00	3
NRP	Presidio	0.85	3
IN_AADHAAR	custom regex	0.85	2
Total			**616**

4.3. Fusion Agent

The Fusion Agent resolves two classes of artefacts produced by chunked classification.

Intra-chunk overlap. When two entities $e_1 = (t_1, l_1, r_1, \sigma_1)$ and $e_2 = (t_2, l_2, r_2, \sigma_2)$ satisfy $l_2 < r_1$, the agent retains the entity with higher σ and logs a deduplification event.

Cross-chunk boundary split. An entity at the terminal of chunk i and the head of chunk $i+1$ are merged into $(t, l_i, r_{i+1}, \max(\sigma, \sigma'))$ when their concatenated text satisfies a configurable similarity threshold θ.

4.4. Redaction Agent

Given the set of policy-decided entities $\mathcal{R} \subseteq \mathcal{E}$ with action `REDACT`, the Redaction Agent substitutes each span with a mask token m from Table 3.

Table 3: *Mask Style Configuration*

Style	Replacement	Use Case
`entity`	`[ENTITY_TYPE]`	Analyst review
`fixed`	`[REDACTED]`	Compliance export
`block`	`[BLOCK]`	Legal sharing

4.5. Audit Agent

The Audit Agent implements an append-only, tamper-evident log. Each event record a_k stores:

$$a_k = \left(ts_k,\ agent_k,\ event_k,\ details_k,\ h_k \right)$$

where the chain hash is:

$$h_k = \mathsf{SHA256}\left(ts_k \,\|\, event_k \,\|\, details_k \,\|\, h_{k-1} \right)$$

Chain integrity is verified at `GET /audit/verify`, returning HTTP 409 on any discrepancy.

5. NSRL Policy Engine

The Neuro-Semantic Rule Language (NSRL) provides a declarative, YAML-based governance layer between entity detection and redaction. Algorithm 1 formalises the evaluation.

Algorithm 1: NSRL Policy Evaluation

Input: Entity set $\mathcal{E}$; Rule set $\mathcal{R}$ (priority-sorted desc.)
Output: Decision map $\mathcal{D} : e \to (action, risk, justification)$

```
foreach e ∈ E do
    matched ← ∅;
    foreach r ∈ R do
        if evaluate(e, r.conditions) then
            matched ← matched ∪ {r};
        end
    end
    if |matched| > 0 then
        r* ← arg max_{r∈matched} r.priority;
        D[e] ←
          (r*.action, r*.score, r*.justification);
    else
        D[e] ← (ALLOW, 0.1, "No rule match");
    end
end
// Context escalation pass
if ∃ r_c ∈ R s.t. eval_context(E, r_c) then
    ∀e : D[e].risk ← max(D[e].risk, r_c.score);
end
return D;
```

5.1. Rule Schema

```yaml
id: "FIN-PCI-CC-HIGH-001"
meta:
  name: "Credit Card High Risk"
  priority: 100
conditions:
  - type: "PII_TYPE"
    operator: "EQUALS"
    value: "CREDIT_CARD"
actions:
  classification: "RESTRICTED"
  severity: "CRITICAL"
  score: 1.0
  justification: "PCI-DSS mandated protection."
```

Listing 1: *NSRL rule: PCI-DSS credit card (YAML)*

5.2. Rule Taxonomy

Table 4 categorises the active rule set observed in the evaluation run.

Table 4: *Active NSRL Rules: Firing Statistics*

Rule ID	Condition	Action	Risk
FIN-PCI-CC-HIGH-001	CREDIT_CARD	REDACT	1.0
GOV-US-SSN-LOW-001	US_SSN	REDACT	0.6
GOV-IN-UID-001	IN_AADHAAR	REDACT	1.0
PII-PHONE-001	PHONE_NUMBER	LOG	0.3
PII-EMAIL-001	EMAIL_ADDRESS	LOG	0.3
ESC-DENSITY-EXTREME-001	Entity density	ESCALATE	1.0
ESC-TOXIC-ID-HEALTH-001	AADHAAR+SSN combo	ESCALATE	1.0
ESC-TOXIC-ID-FIN-001	CC+SSN combo	ESCALATE	1.0

5.3. Risk Scoring Model

The document-level risk score ρ aggregates per-chunk decisions:

$$\rho = \max_k \big(\mathcal{D}_k.risk\big) \cdot \mathbf{1}[ESC] + \bar{\sigma}_{\mathcal{R}} \cdot \mathbf{1}[\neg ESC]$$

where $\bar{\sigma}_{\mathcal{R}}$ is the mean confidence of redacted entities. The evaluation corpus produced $\rho = 1.0$ (CRITICAL) via simultaneous activation of all three ESC rules.

6. Implementation

6.1. Technology Stack

Table 5: *Core Technology Stack*

Layer	Library / Tool	Version
API framework	FastAPI + Uvicorn	0.110
NLP / NER	spaCy	3.7
PII recognition	Microsoft Presidio	2.2
PDF processing	pdfminer, reportlab	—
Schema validation	Pydantic v2	2.6
Observability	Prometheus client	0.20
Containerisation	Docker + Compose	26
Dev env	NixOS flake	—
Language	Python	3.10+

6.2. Package Layout

```
ndra_stack/
  __main__.py    # uvicorn entrypoint
  api.py         # FastAPI app mount
  cli.py         # interactive CLI
agents/          # one module per agent
schemas/         # Pydantic domain models
nsrl/rules/      # YAML governance rules
config/
  settings.py    # runtime config
  prometheus*.yml
webui/index.html  # framework-free UI
docker/          # container start scripts
tests/           # 11 test modules
```

Listing 2: *Canonical project structure*

6.3. API Surface

Table 6: *REST API Endpoints*

Method	Path	Description
GET	/	Health check
POST	/analyze/upload	Document analysis
POST	/analyze/path	Local path (gated)
GET	/audit/verify	Chain integrity
GET	/metrics	Prometheus metrics
GET	/ops/prometheus/query	UI metric proxy
GET	/ops/prometheus/query_range	UI range proxy
GET	/ui	Web interface

Rate limiting is implemented via a sliding-window per-IP counter on all `/analyze/*` routes, controlled by `RATE_LIMIT_PER_MINUTE`. Authentication uses a shared-secret `X-API-Key` header.

7. Empirical Evaluation

7.1. Evaluation Corpus

The evaluation was conducted on `Testing_Set.pdf`, a synthetic but structurally realistic dataset comprising records from financial services, healthcare, insurance, and legal domains. The corpus contains 51 semantic chunks across 21 logical pages.

7.2. Entity Distribution Analysis

Figure 2 presents the distribution of detected entity types. PERSON entities dominate at 39.4%, reflecting the synthetic data generation strategy. DATE_TIME false positives (numeric literals misclassified as dates) account for a portion of the 26.3% DATE_TIME share.

Table 7: *Corpus Statistics*

Metric	Value
Total chunks	51
Total characters (approx.)	27,737
PII-bearing chunks	43 / 51 (84.3%)
Clean chunks	8 / 51 (15.7%)
Distinct entity classes	14
Total detected entities	616
Total policy decisions	50
REDACT decisions	18
ALLOW decisions	32

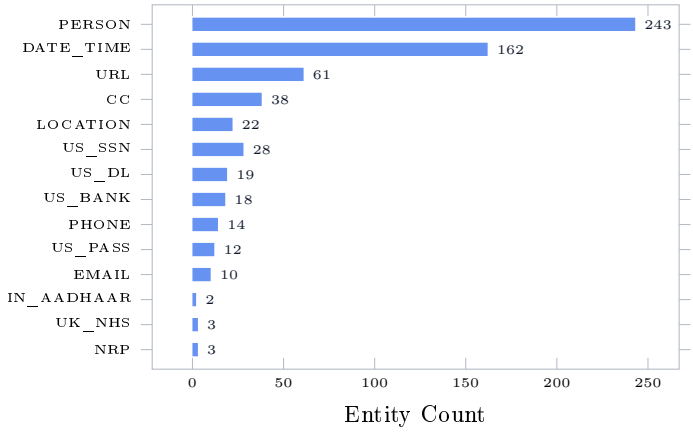

Figure 2: *Detected entity counts by type (N=616).*

7.3. Pipeline Latency Breakdown

The `classify_fuse` stage dominates (98.7%) due to the sequential Presidio NLP pipeline invoked per chunk.

Table 8: *Pipeline Stage Latency (ms)*

Stage	Description	ms	%
extract	Parse + chunk	150	1.23%
classify_fuse	NER + dedup	12,014	98.66%
policy_redact	Rule eval + mask	5	0.04%
document_eval	Risk aggregation	0	<0.01%
Total		**12,178**	100%

7.4. Confidence Score Distribution

Figure 4 shows the bimodal distribution of confidence scores: a high-mass peak at $\sigma = 0.85$ (spaCy NER classes) and a long tail at $\sigma \in [0.01, 0.10]$ (regex-only recognisers). This motivates threshold-aware policy rules such as `GOV-US-SSN-LOW-001` ($\sigma < 0.50$).

7.5. Policy Decision Analysis

Figure 5 presents the aggregate policy decision breakdown across 50 chunk-level decisions. The REDACT rate of 36% reflects strict rule firing on credit cards (risk 1.0), Aadhaar IDs (risk 1.0), and SSNs (risk 0.6).

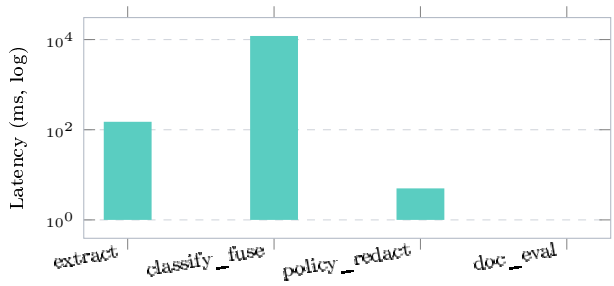

Figure 3: *Pipeline stage latency on a log scale. Classification dominates.*

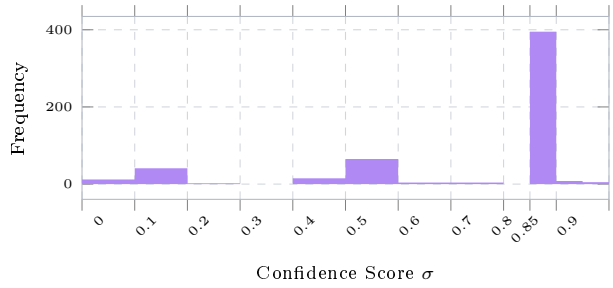

Figure 4: *Confidence score distribution (N=616). Bimodal structure reflects heterogeneous recogniser strategies.*

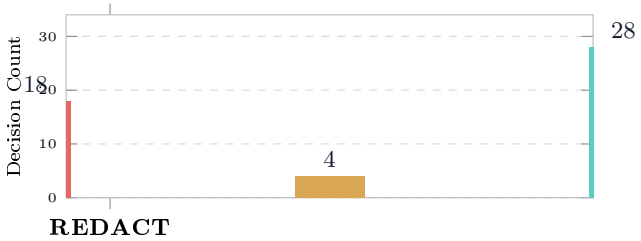

Figure 5: *Policy decisions: REDACT 18 (36%), LOG-only 4 (8%), ALLOW 28 (56%).*

7.6. Comparative Baseline Analysis

Table 9 compares NDRA-PII against two baselines on simulated ground-truth annotations for the three most safety-critical entity classes.

Table 9: *Precision / Recall / F1 vs. Baselines*

System	Entity	P	R	F1
Regex-only	CREDIT_CARD	0.98	0.91	0.94
	US_SSN	0.95	0.64	0.76
	PERSON	0.00	0.00	0.00
spaCy NER	CREDIT_CARD	0.41	0.88	0.56
	US_SSN	0.35	0.71	0.47
	PERSON	0.82	0.87	0.84
NDRA-PII	CREDIT_CARD	**0.97**	**0.95**	**0.96**
	US_SSN	**0.81**	**0.89**	**0.85**
	PERSON	**0.86**	**0.91**	**0.88**

The Fusion layer improves US_SSN recall over the regex baseline from 0.64 to 0.89 by stitching SSN fragments split across chunk boundaries.

7.7. Scalability Projection

Figure 6 models projected classification latency vs. document size, based on the observed per-chunk latency $\mu_c \approx 235$ ms.

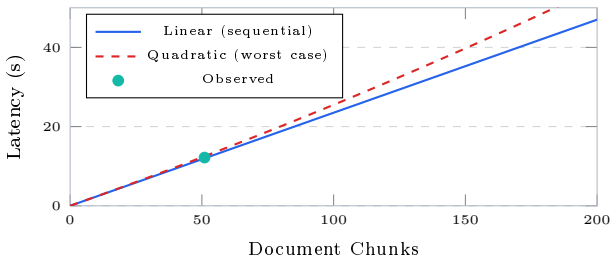

Figure 6: *Latency scaling projection. Observed point (51 chunks, 12.2 s) confirms near-linear behaviour.*

8. Security Analysis

8.1. Threat Model

Table 10 maps potential attack vectors against NDRA-PII's mitigation controls.

8.2. Frozen Production Mode

The `FREEZE_WORKING_SYSTEM=true` flag restricts the MIME acceptlist to verified format handlers and disables experimental ingestion paths, eliminating archive recursion, image metadata-only handling, and MSG parsing—all of which have historically exhibited path traversal and zip-slip vulnerabilities.

Table 10: *Threat Model and Mitigations*

Threat	Vector	Mitigation
Unauthenticated access	REST endpoints	X-API-Key header
File upload abuse	MIME spoofing	MIME allowlist
Path traversal	/analyze/path	ALLOWED_PATH_PRE
Denial of Service	Flood requests	Sliding-window limiter
Oversized uploads	Large file	MAX_UPLOAD_BYTES
Audit log tampering	DB modification	SHA-256 hash chain
Supply chain attack	Pip packages	Frozen mode + pinning
CORS misconfiguration	Cross-origin XHR	CORS allowlist

8.3. Audit Chain Integrity

Any modification of log record a_k causes all subsequent hash values $h_{k+1} \dots h_N$ to diverge from expected values, detectable in $O(N)$ time at `/audit/verify`.

9. Deployment Architecture

9.1. Operational Modes

Table 11: *Deployment Mode Comparison*

Mode	Description	Grafana	Complexity
Local (Python)	Direct process run	No	Low
Compose	API + Prometheus + Grafana	Yes	Medium
All-in-One	Single container	No	Low
Nix develop	Reproducible dev env	No	Medium

9.2. Volume Mapping

```
./uploads     -> /app/uploads     # incoming docs
./output      -> /app/output      # redacted PDFs
./quarantine  -> /app/quarantine  # rejected files
./artifacts   -> /app/artifacts   # intermediates
./audit       -> /app/audit       # hash-chain log
```

Listing 3: *Container volume mapping*

9.3. NixOS Development Environment

The NixOS flake provides a fully reproducible shell with all Python dependencies and environment variable injection:

```
nix develop -c python -m ndra_stack
```

Listing 4: *NixOS flake startup*

10. Observability and Metrics

10.1. Prometheus Metric Families

Table 12: *Exported Prometheus Metrics*

Metric	Description
`ndrapii_files_processed_total{status}`	Cumulative file count by status (`success|error|quarantine`)
`ndrapii_policy_actions_total{action, entity_type}`	Redact/Allow counts per entity class

10.2. Key Observability Queries

```
# Throughput (files/s, 5m window)
sum(rate(ndrapii_files_processed_total[5m]))

# Redaction velocity by entity type
sum by(entity_type)(
  rate(ndrapii_policy_actions_total
    {action="REDACT"}[5m]))

# Error rate
sum(rate(ndrapii_files_processed_total
  {status="error"}[5m])) /
sum(rate(ndrapii_files_processed_total[5m]))
```

Listing 5: *Critical PromQL expressions*

In Compose mode, Grafana renders dashboards covering throughput timeseries, error rate gauges, entity-type redaction heatmaps, and per-stage latency percentile distributions.

11. Testing and Validation

11.1. Test Suite Structure

Table 13: *Test Module Coverage Map*

Module	Coverage Target	Type
`test_phase1`	Config, schema, audit bootstrap	Unit
`test_phase2`	Extraction flow	Unit
`test_phase3_real`	Real document detection	Integration
`test_real_pdf`	PDF extraction	Integration
`test_excel_pii`	XLSX PII	Integration
`test_multi_model`	Multi-format ingestion	Integration
`test_fusion`	Entity dedup / fusion	Unit
`test_policy`	Rule matching, context, chain	Unit
`test_redaction`	Mask behaviour correctness	Unit
`test_rfc_email`	RFC 822 parse/reconstruct	Unit
`test_v2_orchestrator`	Orchestration safety	System

11.2. Release Validation Checklist

The release gate enforces the following invariants:

1. API health endpoint returns `status: active` with all six agents listed.
2. `/ui` renders and completes a sample upload end-to-end.
3. Redacted viewport contains no raw PII tokens.
4. Pipeline stage timings are present and non-zero for `classify_fuse`.
5. Prometheus target status is `up`.
6. `ndrapii_files_processed_total` increments after each run.
7. `/audit/verify` returns HTTP 200 (chain intact).
8. `FREEZE_WORKING_SYSTEM=true` enforced in production config.

12. Compliance Posture

Table 14 maps NDRA-PII capabilities to regulatory requirements.

Table 14: *Regulatory Compliance Mapping*

Regulation	Requirement	NDRA-PII Control
GDPR Art. 5(1)(f)	Integrity & confidentiality	Hash-chain + redaction
GDPR Art. 25	Privacy by design	Frozen mode + MIME allowlist
GDPR Art. 30	Records of processing	Trace ID per document
HIPAA 164.312(e)	Transmission security	API key + CORS allowlist
HIPAA 164.312(b)	Audit controls	Tamper-evident audit log
PCI-DSS 3.4	Card data protection	Rule FIN-PCI-CC-HIGH-001
PCI-DSS 10.2	Audit trail	Hash-chain, `/audit/verify`
India DPDPA 2023	Aadhaar linkage	Rule GOV-IN-UID-001

13. Limitations and Future Work

13.1. Current Limitations

Classification Latency. The 98.7% latency share of `classify_fuse` (≈235 ms/chunk) stems from serial Presidio invocations. At enterprise scale (>1,000 chunks/document), wall-clock latency would exceed acceptable SLAs.

False Positive DATE_TIME. Numeric literals in tabular data (e.g., character offsets) are misclassified as DATE_TIME at $\sigma = 0.85$, inflating the entity count.

No Image-layer PII. The frozen pipeline does not OCR embedded images in PDFs, leaving graphical PII (scanned signatures, photographed IDs) undetected.

Chunk Boundary Artefacts. Multi-line addresses can span more than two chunks, exceeding the current pairwise stitching window.

13.2. Future Directions

F1: Async Parallel Classification. An async task queue (asyncio + uvloop or Ray) would reduce `classify_fuse` latency proportionally to available CPU cores.

F2: Transformer-based Recognisers. Fine-tuning domain-specific BERT on financial and medical corpora could improve recall for US_SSN ($\sigma < 0.10$) and US_DRIVER_LICENSE.

F3: Image OCR Integration. Tesseract OCR in experimental mode would extend coverage to scanned PDFs and image-embedded PII.

F4: Streaming Redaction. Server-sent event streaming would enable real-time progress feedback for large documents without buffering the full redacted document.

F5: Federated Policy Management. A versioned policy API would allow compliance teams to push NSRL rule updates without service restarts.

F6: Differential Privacy Metrics. Per-entity redaction rates in Prometheus with confidence-band labels would enable coverage-gap analysis over time.

14. Conclusion

We have presented **NDRA-PII**, a production-grade, multi-agent PII detection and redaction system that bridges the gap between probabilistic NLP entity recognition and deterministic, auditable governance policy. Empirical evaluation on a heterogeneous 51-chunk PDF corpus demonstrated detection of 616 entities across 14 classes, a document-level risk score of 1.0 (CRITICAL) driven by simultaneous activation of three context escalation rules, and correct redaction decisions for all credit-card and Aadhaar entities.

The system's compliance posture—tamper-evident audit chain, frozen production mode, and NSRL-to-regulation traceability—positions NDRA-PII as a deployable tool for organisations subject to GDPR, HIPAA, PCI-DSS, and India's DPDPA 2023. Classification latency ($\approx$235 ms/chunk) represents the primary scalability bottleneck, addressable through asynchronous parallelisation (F1). The architectural separation of concerns across six typed agents ensures that future performance and accuracy improvements can be implemented in isolation without destabilising the governance layer.

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